Full Stack Assignment – 24

Assignment ID: FSD-24
Title: Support Ticket Management System
Duration: 3 Hours

1️⃣ Objective

Design and develop a **Support Ticket Management System** that allows customers to raise support tickets and support agents to manage them.
The system must enforce reassignment limits and ensure **data persistence using a database**.

2️⃣ User Roles

CUSTOMER

- Register and log in to the system
- Raise support tickets
- View own tickets and their status

AGENT

- Register and log in to the system
- View assigned tickets
- Update ticket status
- Reassign tickets (within limits)

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Ticket Management

- CUSTOMER can create support tickets with issue details
- Tickets must be stored in the database
- AGENT can update ticket status
- Ticket data must be retrieved dynamically

🔄 Ticket Reassignment

- AGENT can reassign tickets to another agent

- Reassignment details must be persisted in the database

4️⃣ Business Rule (Critical)

⚠️ **A support ticket can be reassigned only once.**

Any further reassignment attempts must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All ticket and reassignment data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for CUSTOMER and AGENT
- Ticket creation form for CUSTOMER
- Ticket listing and update view for AGENT

UI design is not evaluated; correctness and flow matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions

- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8 Evaluation Note

Evaluation focuses on **constraint enforcement, role handling, and end-to-end integration**.